

Competency Gap Analysis of Engineering Students: A Systematic Review of Employability Skills, Industry Expectations, and Institutional Practices in India

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ABSTRACT

Engineering Graduate Employability is becoming an increasingly important topic because of the persistent gap in the abilities that are expected by industry and those that are formally educated in engineering programs. Industry now requires rapid adaptation to new technologies, changes in work-place environments, and greater emphasis on soft skills and behavioral competencies than in the past. As such, the required skills for engineering employment are dramatically different than they were several years ago. This review paper reviewed all available literature on competency gap assessment of engineering students, with focus on employability skills, industry expectations, and institutional policies/practices related to competency development in India. Utilizing a systematic review methodology, all peer-reviewed journal articles, policy documents, and institutional publications were identified, collected, evaluated and synthesized to determine which competency gaps exist in four competency categories (technical, soft, managerial, and behavioural). It was found that while engineering programs provide their students with adequate knowledge of theory, there exists a large gap in providing its students with the ability to apply the knowledge learned in a practical manner, the ability to communicate effectively in a variety of contexts, the ability to be adaptable to changing work environments, and professional behaviour. In addition, both industry-academia collaborations and regional disparities contribute to the difficulties of employability issues for many engineering graduates, particularly for those graduating from institutions located in underdeveloped or semi-rural areas. The review indicates significant gaps in future research needs, primarily the need for regionally-based studies and an integration of all stakeholders' perspectives. In summary, this review combines current evidence of employability issues in engineering education, provides recommendations for curriculum revision, policy changes, and future empirical research to reduce the competency gap and improve the employability of engineering graduates.

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1. Introduction

Engineering education has an important role to play in enhancing India's economic growth, technology advancement, and human capital development. Unfortunately, however, a number of national surveys of engineering education and employer studies suggest that most (approximately 80%) engineering graduates lack the competences to be employed at entry-level positions, even though there has been a significant increase in the number of students being enrolled into engineering programs and the number of engineering degrees being awarded [1] [2] [3]. In fact, numerous employer studies reported that only a small percentage of engineering graduates are ready to be employed upon graduation, and the majority require extensive post-graduation training before they become productive employees [1] [2] [4].

The increased demand for engineers possessing technical competence combined with soft skills, higher-order thinking, and digital literacy brought about by rapid digitalization, the advent of Industry 4.0, and globalization further highlights the need for the type of engineers described above [1] [5] [6]. As well as having the ability to perform their job technically, employers now place value on engineering graduates' ability to communicate, work collaboratively with others, solve problems, act professionally, be adaptable, and understand the organizational and social contexts in which they will be working [1] [7] [8] [6]. Despite these changes in what employers expect of engineering graduates, engineering programs continue to be primarily theoretical with minimal opportunities for authentic exposure to the industry; practice-based projects; and competency-based assessments [9] [2] [5] [10] [6].

Research has shown that engineering graduates generally possess sufficient conceptual knowledge, but experience difficulties applying it, communicating effectively in the workplace, and demonstrating appropriate professional behavior, resulting in poor employability rates, underemployment, and high levels of early employee attrition [1] [11] [8] [12] [13]. The problems experienced by engineering graduates are more severe in semi-urban and economically disadvantaged regions, due to college limitations in terms of physical

resources, qualified faculty, connections between the college and the industry, and access to formalized skill development programs [11] [2] [14] [12]. Colleges located in regions that serve rural labor markets and industrial clusters dominated by MSMEs also report lower-quality placement outcomes and fewer high-quality internship experiences [11] [14] [12].

National policy initiatives such as NEP 2020, TEQIP, and Education 4.0/Industry 4.0-reform efforts specifically aim to improve graduate employability through curriculum modernization, outcome-based education, and enhanced industry-educator collaborations [9] [2] [5] [15] [10]. However, implementation is inconsistent and regional-specific constraints and institutional disparities are insufficiently represented in existing empirical research [11] [2] [12] [16]. Present research tends to focus on national averages, metropolitan labor markets, and large companies, with relatively little synthesis of evidence from underdeveloped regions, SMEs, and rural institutions [11] [14] [12] [3].

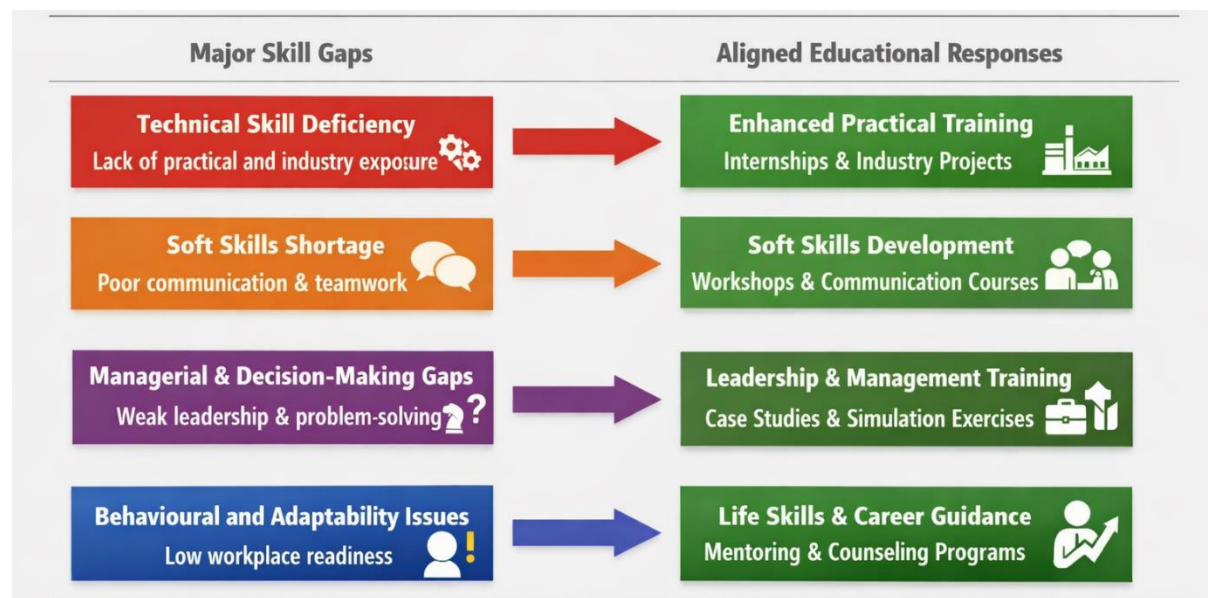


Figure 1: Main empirical reasons motivating this review

Therefore, in light of the need to address both current and emerging issues in engineering education, a systematic review of competency gap analysis in Indian engineering education is both timely and warranted. Through the consolidation of findings related to the employability skills of engineering students, expectations of the industry regarding those same students, institutional practices that impact the preparation of those students, and government policy responses to those challenges, a systematic review of the literature can provide insight into dominant trends, reveal inequities across regions and types of institutions, and highlight areas of the literature that are under researched [9] [2] [4] [6]. This study aims to synthesize existing research on competency gaps of engineering students in India with a focus on variations in

region and institution, and to point out priorities for future empirical research. The goal of this study is to provide evidence-based recommendations for curriculum reform, the establishment of effective educator-industry partnerships, and targeted interventions designed to enhance the professional preparedness of engineering graduates, particularly those graduating from institutions in underdeveloped regions.

Table 1: Key Empirical Motivations for the Review

Motivation for Study	Evidence from Prior Research	Citations
Widespread employer dissatisfaction and skill gaps	Low satisfaction with fresh engineers; only a small share industry-ready	[1] [7] [2] [8] [17] [4]
Theory–practice and soft-skill deficits	Gaps in higher-order thinking, communication, workplace behaviour	[1] [7] [8] [6] [13]
Regional and institutional disparities	Rural/remote colleges, SMEs/MSMEs show weaker linkages and outcomes	[11] [2] [14] [12] [3]
Need to align reforms with Industry 4.0	Education 4.0, learning factories, competency models emerging but uneven	[9] [5] [10] [4] [6]

2. Literature Review

Research on competency gaps and employability of engineering graduates in India consistently highlights a structural misalignment between what institutions teach, what industry expects, and what students actually acquire.

2.1 Industry Expectations and Skill Gaps

Expectations of Industry Regarding Engineering Skills & Perceived Gaps of Skills Surveys involving large numbers of employers of the industry have shown a significant lack of satisfaction regarding the skills of recent graduates from engineering schools: approximately 64% of respondents stated they were "somewhat satisfied" or less satisfied with their new hires [1]. The most important skills for employers are considered to be soft skills (core employability: teamwork, dependability, problem-solving; and communication skills), and these skills are ranked as being the most difficult to find among recent engineering school graduates. Employers also reported that the greatest skill gaps exist in terms of higher level of cognitive skills (analytical, problem-solving, and creative) rather than lower level of cognitive skills (recalling information) [1] [2]. Surveys conducted to determine how employers and institutions perceive employability skills in Deli NCR's IT/Software companies demonstrated that both parties view employability skills as highly important, but each party has a different perception of what constitutes employability skills leading to partial placement opportunities [3]. Likewise, studies focused on MNC software recruitment indicate that there are significant

differences between how graduates perceive themselves and how employers perceive the required skills [4].

Reviews of national and sectorial perspectives support the above mentioned findings. It was estimated that only a very small number of engineers have next generation technical skills (e.g. AI, data engineering, wireless technology) necessary for the changing labor market [5]. Assessments of employability (e.g. AMCAT) have indicated that almost half of all graduates are not employable in any domain of the knowledge economy, while an even smaller percentage are considered "start-up ready" or "design job ready" [6]. The reviews concluded that India is producing too many graduates with weak practical, interdisciplinary and application-oriented competences, resulting in unemployment and under-employment despite the high number of degrees awarded [5] [7] [8].

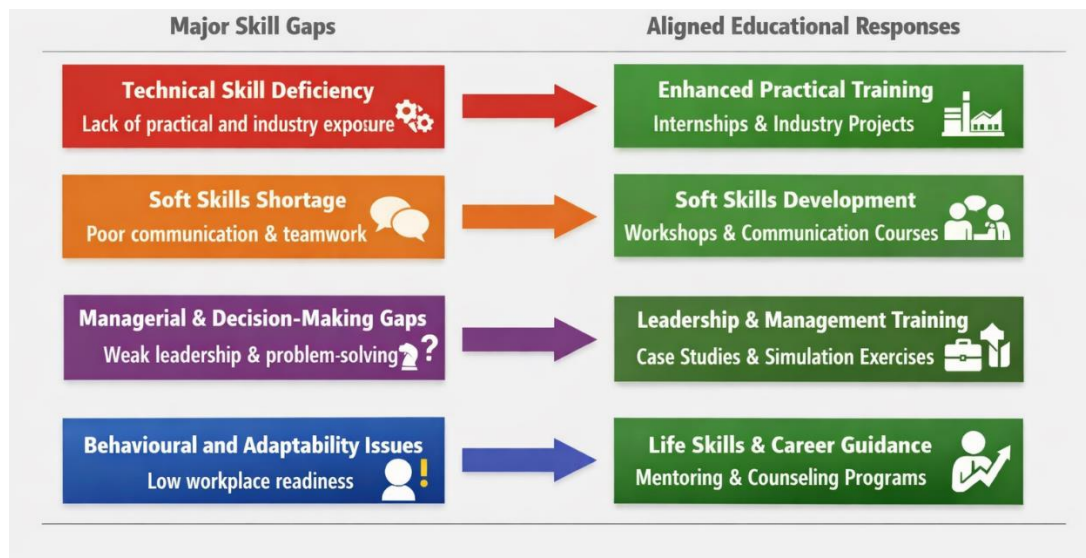


Figure 2: Major skill gaps and aligned educational responses

2.2 Institutional Practices and Curriculum–Pedagogy Issues

There are numerous studies which point out that the competency gap between engineering graduates and their employer's expectations can be attributed to the nature of their curriculum. In particular, the emphasis placed on theoretical, exam-driven mathematics and science is thought to result in a very narrow view of what engineering is. Many of these curricula do not provide much opportunity for engineering students to interact with and learn from industry professionals. It has been reported that only about 40% of engineering students participate in an internship during their time at university. Additionally, most of the students have very little interaction with industry experts and the teaching of engineering subjects is typically through lectures with few opportunities for students to apply the concepts they learn to solve real world problems [5].

Another area of concern is related to communication and language. The case studies indicate that one of the main reasons why many engineering graduates are unemployed is due to their inability to communicate effectively in the workplace. Unfortunately, many of the English language courses that engineering students take are poorly aligned with the needs of potential employers and are delivered in a manner that does not allow for the development of communication based employability [11]. However, there are examples of universities who are trying to address this issue. For example, BITS Pilani has developed courses in business and technical communications that include experiential components such as presentations, group discussions and mock interviews to help students develop soft skills [13].

In addition to communication and language skills, there are also other areas of competence where engineering graduates are found to be lacking. A number of studies have identified significant gaps in the ability of engineering graduates to apply the theories they have learned in class to practical problems. These gaps exist in analytical and problem solving skills, quality engineering knowledge and domain specific knowledge as well as in the ability of engineering graduates to analyze data and make decisions based upon that data [14]. Some researchers believe that a "Learning Factory" could be used to train engineers in the methodological aspects of their profession. The Learning Factory would provide students with a simulated industrial environment in which they could gain experience in applying the theories they had learned in class to real world problems. This type of experience would significantly enhance the methodological competence of engineering graduates and reduce the need for costly graduate training [15].

2.3 Underemployment, Skill Mismatch, and Structural Factors

Research studies have shown that despite being employed as engineers, many engineering graduate students perform tasks and hold jobs that are beneath their qualifications and education level, which is attributed to an over-qualified education for the job at hand; outdated curriculum content; weak links between academia and the private sector/industry; lack of direction/career counseling; and/or lack of experience/limited opportunities for practical learning/training prior to graduation [7]. However, studies show that engineering graduate students who participate in internship or skill development programs have less chance of being underemployed after graduation, and support the need for practice based, and industry aligned training programs [7]. Furthermore, research reviews of the broader employability environment highlight that employers require and want more than just a broad base of knowledge and skills from graduates, but also entrepreneurial orientation, however, the institutions response has been very uneven [18] [8].

2.4 Emerging Responses and Promising Practices

A number of proposals and examples of institutional and policy responses have been made across the literature, including:

- a) Curriculum reform, that is, to emphasize higher order thinking and to combine employability into the engineering curriculum (that is, to teach problem solving, design and creativity and to treat knowledge and professional skills as being interdependent as opposed to separate) [1] [19] [2].
- b) Project based and experiential learning: there are a number of studies that provide evidence that experiential “Live-in-Labs” and structured competitive programming can provide a high level of correlation between course outcomes and program outcomes that relate to employability, and improve students’ ability to solve real world problems in teams, create new solutions through innovation, etc. [6].
- c) Increased linkage with industry: it has been suggested that increased linkage with industry can be achieved by having faculty visit industry regularly; providing students with longer internships; using capstone projects to address real industrial problems; and conducting systematic reviews of the curriculum by faculty, alumni, and industry to ensure that the competencies taught are relevant to current industry needs [18] [15] [17].
- d) Specifically targeted communication and soft skill interventions: It has been suggested that improvements in communication and soft skills for graduates can be achieved through targeted interventions, including the development of activity-based communication courses; English for Specific Purposes / English for Employability (ESP/EE) approaches; and improved teacher education for language teachers [13] [11] [12].

Table 2: Competency Gaps and Institutional Responses

Dimension	Main Gaps Identified	Suggested / Observed Practices	Citations
Soft & communication skills	Poor communication, teamwork, confidence; employer–student misperceptions	Activity-based communication courses, ESP, soft-skills modules	[1] [13] [4] [3] [11] [12]
Higher-order cognition	Weak problem solving, analysis, creativity	Problem-/project-based learning, competitive programming	[1] [14] [6] [19] [2]
Application & methodological skills	Theory–practice gap, limited industry exposure, methodological deficits	Learning factories, internships, industry projects, quality-based redesign	[5] [15] [14] [9] [16] [17] [10]

Dimension	Main Gaps Identified	Suggested / Observed Practices	Citations
Next-gen technical skills	Low AI, data, wireless, interdisciplinary skills	Curriculum modernization, industry-driven competency models	[7] [5] [9] [17] [10]

3. The Competency Gaps in Graduates from Engineering Programs

There is evidence to support the fact that competency gaps in engineering graduates exist in multiple dimensions and have continued to be present over time. The gaps found were technical, behavioral, managerial, organizational and soft in nature; all of these gaps had an impact on the employability of the graduates, their performance at work and their ability to sustain themselves throughout their careers. The following sub-sections will describe the competency gaps found in each of the areas studied.

3.1 Technical Skill Gaps

3.1.1 Mismatch Between Theory and Practice

Many of the studies completed in order to evaluate the competency gaps of engineering graduates, cite a significant mismatch between the theoretical knowledge gained through engineering programs and the practical application of that knowledge. While there is emphasis placed upon gaining theoretical knowledge of subjects and concepts in engineering programs, the graduates of these programs often do not know how to take the knowledge gained and apply it to industrial problems in the same manner as their counterparts working in the industry. In addition to this, the laboratory experiences and academic projects completed by students while they are enrolled in engineering programs are typically not aligned with the contemporary practices, tools and technologies used within the industry.

Due to the theory-focused nature of engineering programs, students are not able to connect their academic learning to functional workplace competencies. This has resulted in employers requiring extensive on-the-job training in order to develop the necessary workplace competencies, resulting in increased costs associated with inductions and decreased levels of productivity in the first year after graduation.

3.1.2 Limited Industry Experience

Another important factor that contributes to technical skill gaps in engineering graduates is the limited amount of experience they have with industry. Many engineering graduates do not participate in internships, live projects or other forms of structured industrial training. Due to this limited experience, the graduates do not have an opportunity to become familiar with the day-to-day operations of industry including, but not limited to, professional workflows, production systems, quality standards and workplace technologies.

In addition to limiting the graduates' ability to understand the practical constraints of industry and the way in which problems are solved in an applied setting, the lack of direct experience in industry also limits the graduates' level of technical preparedness and confidence in their professional roles.

3.2 Soft Skill and Behavioral Gaps

3.2.1 Deficiencies in Communication Skills

Deficiencies in communication skills are consistently reported as one of the most serious weaknesses in engineering graduates. This includes, but is not limited to, deficiencies in verbal communication, technical writing, presentation skills and interpersonal interaction. Students graduating from engineering programs typically have difficulty expressing their ideas clearly, engaging in professional discussions and interacting in a collaborative team environment.

As a result of these communication deficits, graduates have difficulty being selected for jobs, collaborating with colleagues and developing professionally despite having sufficient technical knowledge.

3.2.2 Lack of Adaptability and Professionalism

Behavioral competencies that are becoming increasingly important to employers, such as adaptability, professionalism, work ethic and emotional intelligence, are also lacking in many engineering graduates. In addition to experiencing difficulties adapting to the organizational culture of their employer, many graduates also experience difficulty receiving constructive feedback and demonstrating a level of professionalism expected by employers.

These deficits often manifest during the new employee phase of employment, creating confusion regarding job responsibilities, workplace-related stress and interpersonal conflicts that may eventually lead to job dissatisfaction and early job abandonment.

3.3 Managerial and Organizational Skill Gaps

3.3.1 Ability to Make Decisions

Engineering graduates typically exhibit limited decision making abilities, particularly in complex or uncertain environments. As engineering programs provide little opportunity for students to make independent judgments, assume leadership roles and solve problems in a responsible manner, graduates are inclined to rely on instruction to guide them in their decision making processes.

As a result of the inability of engineering graduates to make decisions independently, there is a negative effect on efficiency, accountability and leadership potential in professional settings where decisions must be made quickly and efficiently, particularly in project based roles.

3.3.2 Ability to Manage Time and Stress

Time management and stress management are essential organizational skills required in today's workplace. Unfortunately, many engineering graduates lack the ability to manage their time effectively and handle the pressures of a performance driven workplace. Unlike academic schedules, which are typically highly structured and provide limited exposure to the intensity of the real world workload and accountability of their future employer, graduates often find themselves ill-prepared to manage their time and stress effectively in their professional careers. When engineers are unable to manage stress effectively in the workplace, it can have a negative impact on their productivity, mental health and overall career sustainability.



Figure 3: Identified Competency Gaps in Engineering Graduates

4. Conclusion

This review paper examines the current literature regarding competency gap assessment of engineering graduates in India, specifically the employability skills of engineering graduates, the expectations of the Indian industry, the practices of higher education institutions, and the differences between various regions in India. The synthesis of empirical research, policy documents, and institutional data demonstrate that while there have been significant expansions in the number of engineers graduating in India, many engineering graduates still exhibit substantial competency gaps that negatively impact their employability and early career success.

The review confirms that engineering graduates generally have a solid theoretical base of knowledge, but they often cannot utilize this knowledge in an industrial setting. The largest gaps identified were the theory–practice gap, too little industrial experience, poor communication skills, limited ability to be adaptable and professional, and poor managerial and organizational competences such as decision-making, time management, and managing

stress. The collective effect of these competency gaps affects the workplace preparation of engineering graduates, their productivity in employment, and ultimately their long-term career viability.

The study indicates that the employability issues faced by engineering graduates do not exist uniformly throughout India. Engineering graduates from less developed and semi-urban areas face greater disadvantages than those from urban areas, primarily because of fewer institutional resources, fewer links to the industry, and less opportunity to gain hands-on experience. While national level initiatives, such as Outcome-Based Education (OBE), Industry 4.0 Reforms, and Skill Development Policies are intended to help mitigate some of these problems, the implementation of these programs has been inconsistent and does not consider the regional and institutional contexts.

There is a need for additional research to fill in the gaps in the current literature. Many prior studies examine competency gaps based upon national trends or metropolitan labor market conditions, few studies provide empirical analyses of competency gaps specific to a given region, and none of the prior studies integrated perspectives of students, faculty members, recruiters, and policymakers into their analyses. Therefore, there exists a significant need for future research using a combination of qualitative and quantitative methods to investigate how competency gaps develop over time and across regions.

In summary, reducing the competency gap in engineering education will require a multifaceted and collaborative strategy that includes curricular revisions, experiential and project-based learning, behavioral and social skill interventions tailored to individual needs, and ongoing collaboration between academia and industry. Educational policymakers and institutions must give priority to developing and implementing contextually sensitive strategies that reduce the disparities in competency gaps between different regions of India and meet the changing needs of employers. This review integrates prior research to create a foundation for future empirical research and to guide educational reforms designed to enhance the employability and job-readiness of engineering graduates in India.

5. Conflict of Interest

The author(s) declare that there is **no conflict of interest** regarding the publication of this paper

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